

Josh Noll

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Summary

Cloud/DevOps Engineer specializing in Kubernetes, infrastructure automation, and secure cloud platforms supporting enterprise and mission environments. Active TS/SCI clearance; MA in IT Management; CISSP, SecurityX, CCNA certified. Driven to build durable solutions to long-standing pain points, bridging engineering and leadership to deliver measurable outcomes.

Skills

Azure | AWS | Linux | PowerShell | Bash | Ansible | Python | HTML | CSS | JavaScript | TypeScript | React | Vite | NextJS | Jest | Tailwind | Docker | Kubernetes | Helm | FluxCD | ArgoCD | Temporal | VMWare | Proxmox | Wireguard | Tailscale | Nginx | Traefik

Certifications

[CISSP](#) | [CCNA](#) | [Az-104](#) | [Az-900](#) | [SecurityX](#) | [Security+](#) | [Network+](#) | [Project+](#)

Experience

Infrastructure Engineer, Unstructured (*Feb 2026 – present*)

- Designed and delivered versioned, immutable dedicated-instance infrastructure configuration using Kubernetes CRDs, Pulumi, Temporal, admission controls, and operator reconciliation; enabled teams to canary changes on individual instances before fleet-wide rollout.
- Built and productionized an Internal Docs platform with Docusaurus, FastAPI, Helm, Docker, Okta OIDC, local search, CI/CD, and an authenticated MCP server; migrated 189 pages to Git-backed durable storage and automated updates across 199 repository sessions.
- Diagnosed and fixed production-scale operator-console performance bottlenecks, reducing API payloads by 97.6% and 91% and bringing all seven tested navigations to 323 ms or less.
- Removed EKS upgrade blockers across 21 AWS clusters by making Kubernetes versions configurable, hardening cross-cloud validation, and pinning managed node groups to the control-plane version.
- Authored 162 pull requests across 11 repositories, with 140 merged, and reviewed at least 297 pull requests across infrastructure, release engineering, internal documentation, and developer tooling.
- Completed the infrastructure ingress migration from Nginx to Traefik, including removal of obsolete controllers and resources.
- Built and operationalized an AI engineering orchestrator with automated issue workers, CI/PR monitoring, re-review after blocker fixes, an append-only event ledger, a live dashboard, and a terminal-native TUI; production validation included 321 passing tests.
- Led a platform release through late-stage failures, halted an unsafe promotion, corrected staging/production plugin divergence, reran production promotion, and confirmed end-to-end validation before release.
- Assessed a 23-instance infrastructure fleet using live status and deployment previews, separating harmless refresh drift from material backlog and creating a risk-based maintenance and migration plan.
- Made infrastructure drift state self-healing after deployments by updating the operator to re-check and clear stale verdicts immediately rather than waiting up to 24 hours.
- Created authenticated, live documentation previews for every content pull request, removing the need for reviewers to run Docusaurus locally.

Data and Information Systems Engineer, US Army (*Jan 2021 – Jan 2026*)

- Delivered an AI-powered search solution using Azure AI Search with cross-functional teams, improving accessibility of 22TB+ of data and cutting information retrieval time from 20+ minutes to under 10 seconds.
- Identified a chronic gap in user account management and built a custom PowerShell module from scratch, increasing service desk ticket closure speed by 100%.
- Authored an enterprise cybersecurity policy aligning technical controls with business priorities; secured C-suite buy-in and reduced violations by 75%+.
- Designed and presented a tactical network/systems architecture to 5+ C-suite stakeholders, enabling buy-in for a training event supporting 50+ end users.

- Deployed resilient Active Directory domain architecture across 8 geographically dispersed sites, maintaining service continuity for 150+ users in austere tactical environments.

Executive Assistant, US Army (*Feb 2020 – Jan 2021*)

- Bridged C-suite leadership and technical teams in drafting and publishing the U.S. Army Arctic Strategy, a foundational initiative driving the Army's return to Arctic dominance amid great power competition.

Platoon Leader, US Army (*Mar 2019 – Nov 2020*)

- Commanded a 22-person Cavalry platoon through a successful combat training center rotation; rated #1 of all platoon leaders in the squadron and accountable for \$1M+ in equipment.

Volunteering

Programming Director, Arrowhead Research (*Jan 2026 – present*)

- Lead Black Relay 2026: Safehouse Systems, a four-month engineering initiative to build a portable, survivable operating stack for temporary sites across fabrication, detection, survivability, and TAK teams.
- Designed, built, and deployed the Arrowhead Research website and the Black Relay 2026 program site, translating the organization's mission, programs, application flow, and field-event requirements into public-facing web experiences.
- Delivered a presentation at a company-sponsored prototype showcase on open-source methodology to a panel of 5 industry experts and DoD leaders including the US Army CTO, demonstrating technical leadership and communication skills.

Web Developer, Cybersecurity Canon (*Nov 2025 – May 2026*)

- Prototyped a modern platform to replace long-deferred WordPress infrastructure; communicated architectural trade-offs to leadership to enable enhanced extensibility.

Projects

Black Relay 2026: Safehouse Systems [↗](#)

Designed and deployed the public program site for a four-month field-engineering initiative building a portable, survivable safehouse operating stack for temporary sites.

Homelab as Code [↗](#)

GitOps/IaC platform deploying homelab services to Kubernetes on Talos Linux; uses FluxCD, Tailscale, SOPS, and age for secure multi-user operations and secret management.

Homelab Ansible Collection [↗](#)

Published Ansible Galaxy roles that automate Linux server hardening, Docker installation, containerized service deployment, Tailscale networking, Traefik reverse proxying, and Cloudflare DNS records.

Personal Website [↗](#)

Astro-based personal site deployed through GitHub Actions to Azure Storage and secured behind Cloudflare HTTPS, with a technical blog and public notes.

pi-teach [↗](#)

Published Pi extension that turns agent sessions into interactive HTML classrooms with highlight-to-ask explanations, persistent contextual Q&A, and agent-graded quizzes.

Traefik Install Playbook [↗](#)

Ansible playbook that automates Traefik deployment with Tailscale and Traefik Kop across multiple hosts.

Education

Webster University: MA, Information Technology Management (*May 2022*)

University of Wisconsin - Eau Claire: BA, German (*May 2017*)

Galvanize: Software Development Immersive (*September 2025*)

US Army Cyber Center of Excellence (CCOE): Graduate, Information Systems Engineering Course (*October 2021*)